SCHOOL OF COMPUTATION, INFORMATION AND TECHNOLOGY — INFORMATICS

TECHNISCHE UNIVERSITÄT MÜNCHEN

Bachelor's Thesis in Informatics

Bidirectional Morphological Inflection Generation and Analysis

Murilo Escher Pagotto Ronchi

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Bidirektionale morphologische Flexionsgenerierung und -analyse

Author:	Murilo Escher Pagotto Ronchi
Examiner:	Shu Okabe Ph.D.
Supervisor:	Shu Okabe Ph.D.
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I confirm that this bachelor's thesis is my own work and I have documented all sources and material used.

Munich, February 27, 2026 Murilo Escher Pagotto Ronchi

Acknowledgments

Abstract

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1 Introduction

Morphological systems encode how words are formed and how their forms vary with grammatical function. For languages with rich morphology, a single lemma may correspond to a large paradigm of inflected forms, and the mapping between form and meaning can involve non-trivial orthographic and phonological alternations. Robust handling of morphology is therefore central to a wide range of downstream natural language processing (NLP) applications, including syntactic parsing, information extraction, and machine translation.

This thesis focuses on two core tasks that operationalize morphological competence: *morphological inflection* and *morphological analysis*. In the inflection task, the system receives a lemma together with a bundle of morphological features and must generate the corresponding inflected surface form. In the analysis task, the system must recover the lemma and the morphosyntactic description (MSD) from an observed inflected form. In addition to the form alone, many modern settings provide sentence context, which can disambiguate morphological analyses that are underspecified at the token level.

The main goal of this work is to analyze how different modeling approaches perform on these tasks across multiple typologically diverse languages, and to understand how data sources and annotation conventions affect the interpretation of results. Concretely, we evaluate classical and neural baselines alongside modern sequence-to-sequence models and prompting-based large language model (LLM) baselines.

Models. For morphological inflection, we compare (i) a non-neural SIGMORPHON baseline, (ii) a neural transducer baseline, and (iii) fine-tuned ByT5 models. For morphological analysis, we evaluate ByT5 models trained in the inverse direction on UniMorph, as well as context-based experiments on Universal Dependencies (UD)-derived data, including a context-aware ByT5 variant and inference-only LLM baselines. Finally, we include a bidirectional ByT5 model that is trained jointly on forward (inflection) and inverse (analysis) examples, enabling a unified investigation of multi-task learning for both mappings.

Datasets and evaluation. The experiments draw on two widely used morphological resources: UniMorph (Batsuren et al., 2022) and Universal Dependencies (Nivre et al., 2020). Because these resources differ in annotation schemes and file formats, we construct unified train–development–test splits and a common instance representation for all model families. For the UD-based setting, we map UD feature bundles into a UniMorph-style tag inventory using an official conversion tool (McCarthy et al., 2018) where translators are available. When translators are missing, UD features cannot be converted into a comparable UniMorph-style MSD inventory, which has direct implications for which metrics can be reported consistently across languages.

Contributions. This thesis makes the following contributions:

- It provides a unified and reproducible experimental pipeline for training and evaluating a diverse set of models on morphological inflection and analysis across multiple languages.
- It offers a systematic comparison between established shared-task baselines, fine-tuned ByT5 models, and prompting-based LLM baselines under consistent preprocessing and evaluation.
- It highlights how dataset properties and annotation conventions, including differences between UniMorph and UD and language-specific lemma conventions, can create apparent outliers and limit cross-resource comparability.

Thesis outline. Chapter 2 introduces the tasks and the baseline methods. Chapter 3 describes the datasets, languages, and preprocessing steps used to construct unified splits. Chapter 4 details the experimental setup, model configurations, and evaluation procedure. Chapter 5 presents and discusses the results for inflection and analysis. Finally, Chapter 6 summarizes the main findings, discusses limitations, and outlines directions for future work.

2 Background

Different models have been used in this thesis to tackle two different, but related tasks. This chapter presents both tasks, the theoretical background and functionality behind the models used, as well as explanations of linguistic concepts pertaining to the tasks.

2.1 Tasks

This thesis addresses two closely related problems that can be viewed as inverse linguistic operations. Both tasks, introduced in the following subsections, involve applying a set of linguistic transformations to a base form of a word in order to produce an inflected form.

A lemma, commonly referred to as the canonical form, denotes the base representation of a word. This base form can undergo various transformations belonging to different grammatical categories, including mood, aspect, tense, case, number, and person. These transformations are represented as a collection of morphological features, which in this work are often referred to as *morphological tags*. Applying a specific set of such features to a lemma yields the corresponding inflected form.

2.1.1 Morphological Inflection

Morphological inflection refers to the process of generating a surface word form from a canonical base form (lemma) and a set of grammatical properties. Given a lemma and features describing tense, number, person, case, or other morphosyntactic categories, the task consists of producing the corresponding inflected form. An example in English is given in Figure 2.1.

$$(\textit{walk}, \{\textsc{verb}, \textsc{past}, \textsc{third person}, \textsc{singular}\}) \;\rightarrow\; \textit{walked}$$

Figure 2.1: Illustration of the morphological inflection task. Given a lemma and a set of grammatical features, the goal is to generate the corresponding inflected form.

Automatic morphological inflection has long been studied in computational linguistics. Early approaches relied on rule-based systems and finite-state transducers (Koskenniemi, 1983; Kaplan and Kay, 1994; Beesley and Karttunen, 2003), which required curated rules determined by linguistic experts. More recently, the problem has been formulated as a supervised sequence-to-sequence learning task, in which neural encoder-decoder models operate directly on character sequences (Faruqui et al., 2016; Kann and Schütze, 2016; Aharoni and Goldberg, 2017). Such models learn morphological transformations from data and have become the dominant paradigm for the task.

The task has gained particular prominence through a series of multilingual shared tasks, most notably the SIGMORPHON and CoNLL–SIGMORPHON reinflection challenges (Cotterell et al., 2017; Cotterell et al., 2018; Goldman et al., 2023). These benchmarks provide standardized training and evaluation settings across dozens of typologically diverse languages and have established morphological inflection as a common testbed for studying character-level generalization and low-resource learning. Large-scale resources such as UniMorph further facilitate this research by offering broad cross-linguistic coverage and consistent morphological annotations (Batsuren et al., 2022). The UniMorph annotation style encodes a MSD as a single string of semicolon-separated feature labels (e.g., `V;PST;3;SG`). In other words, the set of grammatical properties in Figure 2.1 is represented as a compact sequence of tags rather than as a structured set. Using this notation, the example in Figure 2.1 can be written as

$$(\textit{walk}, \{\texttt{V;PST;3;SG}\}) \;\rightarrow\; \textit{walked}.$$

2.1.2 Morphological Analysis

Full morphological analysis refers to the task of recovering both the lemma and the associated set of morphological descriptors from an inflected surface form. The task is composed of the subtasks of lemmatization and morphological tagging, which predict the canonical form and grammatical features, respectively. An example in English is shown in Figure 2.2.

$$\textit{walked} \;\rightarrow\; (\textit{walk}, \{\textsc{verb}, \textsc{past}, \textsc{third person}, \textsc{singular}\})$$

Figure 2.2: Illustration of the morphological analysis task. Given an inflected form, the goal is to predict the corresponding lemma and set of grammatical features.

Automatic morphological analysis has traditionally been addressed using rule-based and finite-state techniques similar to those employed for inflection. Classical two-level and finite-state morphology frameworks model analysis and generation within a single bidirectional system, enabling both the decomposition of surface forms into lexical representations and the production of inflected forms from abstract specifications (Koskenniemi, 1983; Beesley and Karttunen, 2003).

More recently, data-driven approaches have largely replaced manually engineered analyzers. Related problems are often formulated as lemmatization or morphological tagging, where models predict either the lemma or the set of grammatical features directly from the surface form. Neural character-level architectures have proven effective for these tasks, learning morphological regularities directly from data without explicit rules (Malaviya et al., 2019; Cotterell and Heigold, 2017). In this thesis, we adopt the more general formulation of full morphological analysis, which *jointly* predicts both the lemma and the complete set of morphological descriptors.

Beyond these task formulations, accurate morphological modeling plays an important role in downstream natural language processing applications more broadly. Generating and analyzing correct surface forms is essential for systems that must produce grammatically well-formed text, such as machine translation and text generation, and for tasks that rely on morphological features, including tagging and syntactic parsing. Prior work has shown that incorporating morphological or subword structure improves neural machine translation for morphologically rich languages (Sennrich et al., 2016), while morphology-aware representations and feature prediction benefit language modeling and tagging performance (Botha and Blunsom, 2014; Cotterell and Heigold, 2017). These findings motivate the study of both inflection and analysis as complementary components of robust morphological processing.

2.2 Morphological Inflection Baselines

2.2.1 Non-Neural Baseline

Earlier approaches to morphological inflection primarily relied on explicit hand-crafted transformations to generate inflected word forms. In line with this tradition, the first baseline evaluated in this work adopts a simple edit-rule strategy.

The system is provided as part of the SIGMORPHON 2023 Shared Task (Goldman et al., 2023). It builds upon a baseline introduced in the SIGMORPHON 2020 edition (Vylomova et al., 2020) and is based on the model originally proposed by Cotterell et al. (2017).

Unlike neural sequence-to-sequence approaches, this model does not learn continuous representations. Instead, it derives explicit string edit rules directly from the training data. For each training triple consisting of a lemma, a set of morphological tags, and the corresponding inflected form, the system first aligns the two strings using Levenshtein distance. The aligned pair is then decomposed into a prefix part, a shared stem, and a suffix part. Based on this decomposition, both prefix-changing and suffix-changing transformation rules are extracted.

To illustrate this procedure, we reproduce a simplified alignment example adapted from the official baseline documentation (Vylomova et al., 2020). For the German pair *schielen* $\rightarrow$ *geschielt* ("to squint"), one possible alignment is:

```
__schielen
geschielt_
```

which can be segmented into prefix (Pr), stem (St), and suffix (Su) regions:

```
Pr  St       Su
__  schiele  n
ge  schielt  _
```

From this decomposition, the system extracts both prefix rules (e.g., $\epsilon \rightarrow$ *ge*) and suffix rules describing the stem-final changes. Here, ϵ represents an empty string.

Each training example may therefore yield multiple candidate rules of varying specificity. During inference, the system first selects the rule whose left-hand side matches the longest portion of the lemma (suffixes for suffix rules or prefixes for prefix rules). If several rules of equal length apply, the most frequent one observed in the training data is chosen.

Table 2.1 illustrates representative examples of such transformations. For clarity, the examples focus on suffix edits, which are more common in English, although the same procedure is applied analogously to prefixes. By aggregating such examples across the training data, the model learns which modifications are most likely for a given morphological specification.

Importantly, each training pair contributes multiple candidate rules rather than a single transformation. For example, from the pair *walk* $\rightarrow$ *walked*, the alignment procedure yields a hierarchy of suffix edits with increasing specificity, including $\epsilon \rightarrow$ *ed*, *k* $\rightarrow$ *ked*, *lk* $\rightarrow$ *lked*, *alk* $\rightarrow$ *alked*, and *walk* $\rightarrow$ *walked*. This strategy encourages more specific transformations while still allowing general rules to apply when appropriate.

While this allows the model to efficiently capture regular patterns, it can lead to overgeneralization when training data is sparse. For instance, if the only observed pair were *speak* $\rightarrow$ *spoke*, the model could learn the rule *eak* $\rightarrow$ *oke*. Applying this rule to an unseen lemma such as *leak* would incorrectly yield *loke*, instead of *leaked*. This example illustrates how purely surface-based edits may fail to generalize appropriately.

Despite these limitations, the approach provides a simple, transparent, and computationally efficient baseline. Its explicit rule extraction makes the predictions easily

Lemma	MSD	Inflected form	Example candidate rule
walk	V;PST;3;SG	walked	$\epsilon \rightarrow$ ed
talk	V;PST;3;SG	talked	lk $\rightarrow$ lked
play	V;PST;3;SG	played	ay $\rightarrow$ ayed
jump	V;PST;3;SG	jumped	mp $\rightarrow$ mped
call	V;PST;3;SG	called	all $\rightarrow$ alled
love	V;PST;3;SG	loved	ve $\rightarrow$ ved
move	V;PST;3;SG	moved	ove $\rightarrow$ oved
try	V;PST;3;SG	tried	y $\rightarrow$ ied
stop	V;PST;3;SG	stopped	p $\rightarrow$ pped
fix	V;PRS;3;SG	fixes	$\epsilon \rightarrow$ es

Table 2.1: Illustrative examples of edit rules extracted by the non-neural baseline. The table focuses on suffix rules for readability, although both prefix and suffix transformations are learned.

interpretable and offers a useful reference point for comparing the more complex neural models introduced in the following sections.

2.2.2 Neural Baseline

The other reference system used is the official neural baseline provided as part of the SIGMORPHON 2023 Shared Task. This system corresponds to the character-level Transformer model proposed by Wu et al. (2021), which has been adopted as a strong baseline for morphological inflection.

The model follows a standard encoder-decoder Transformer architecture (Vaswani et al., 2017), operating directly on character sequences. To better suit low-resource transduction tasks, a compact configuration is used, consisting of four encoder layers and four decoder layers with multi-head self-attention. Each layer employs four attention heads, a model dimension of $d_{\text{model}} = 256$, and a feed-forward dimension of $d_{\text{FF}} = 1024$. Decoding is performed autoregressively. For feature-guided tasks such as morphological inflection, morphological tags are encoded as special feature tokens that are concatenated with the input character sequence and processed jointly by the encoder. Wu et al. (2021) further propose a feature-invariant positional encoding scheme to avoid imposing an arbitrary order over the feature set.

2.3 ByT5

In addition to character-level and token-based models, the other analyzed model is ByT5 (Xue et al., 2022), a token-free variant of the T5 architecture (Raffel et al., 2020). ByT5 is an encoder-decoder Transformer model that operates directly on UTF-8 byte sequences instead of subword tokens. Unlike conventional Transformer models that rely on learned vocabularies (e.g., SentencePiece), ByT5 uses a fixed vocabulary of 256 byte embeddings corresponding to all possible byte values, thereby eliminating the need for tokenization.

Architecturally, ByT5 follows the T5 framework and its multilingual variant mT5 (Xue et al., 2021). To better accommodate longer byte sequences, the encoder and decoder depths are decoupled, employing a substantially heavier encoder stack with a 3:1 encoder–decoder ratio, in contrast to the balanced configuration used in T5 and mT5.

Because ByT5 operates directly on raw byte sequences, it has been hypothesized to be particularly suitable for tasks that are sensitive to orthographic or phonological structure, including morphological inflection. Xue et al. (2022) provide the performance results of ByT5 on the SIGMORPHON 2020 (Vylomova et al., 2020) tasks of grapheme-to-phoneme conversion and morphological inflection as part of their benchmark. This property makes it especially relevant for the evaluation conducted in this thesis.

2.4 Large Language Models

Large language models (LLMs) are Transformer-based neural networks pretrained on large-scale text corpora using self-supervised objectives. Owing to their scale and broad pretraining, such models exhibit strong generalization capabilities and can perform a wide range of sequence-to-sequence tasks with little or no task-specific training. Since both morphological inflection and morphological analysis can be formulated as text generation problems, LLMs provide a natural evaluation framework without dedicated fine-tuning.

In this work, we evaluate two open-weight instruction-tuned LLMs: Qwen3-8B (Yang et al., 2025) and Llama-3.1-8B-Instruct (Grattafiori et al., 2024). Both models are decoder-only Transformer architectures with approximately 8 billion parameters and are designed for general-purpose text generation. The 8B parameter scale represents a compromise between strong performance and practical computational requirements.

Qwen3-8B, developed by Alibaba, is part of the Qwen model family and is trained on large-scale multilingual data to provide strong general-purpose language understanding and generation capabilities. Llama-3.1-8B-Instruct, developed by Meta AI, belongs to the

LLaMA series and incorporates improved pretraining data and alignment procedures relative to earlier versions. Both models are instruction-tuned to follow natural language prompts, making them suitable for prompt-based evaluation settings.

3 Datasets and Languages

3.1 Datasets

To address the tasks of morphological inflection and morphological analysis and to train the proposed models, labeled data is required. Each instance consists of a lemma, a set of morphological tags describing the target grammatical features, and the corresponding inflected form. For this purpose, this work relies on two widely used resources: UniMorph (Batsuren et al., 2022) and UD (Nivre et al., 2020).

3.1.1 UniMorph

UniMorph is a multilingual project dedicated to providing large-scale morphological resources in the form of inflection tables and structured morphological annotations across a wide range of languages. At the time of writing, the dataset covers up to 182 languages, including both high-resource and low-resource as well as endangered languages.

For each language, UniMorph provides a dataset consisting of individual instances that pair a lemma with a bundle of morphological features and the corresponding inflected form. Concretely, the data is distributed as tab-separated lines, where each row represents a single lemma, tags, form triple. Multiple rows typically share the same lemma and correspond to different morphological feature combinations, effectively encoding the full inflectional paradigm in a flat, line-based format, as illustrated in Table 3.1.

Although the most recent UniMorph publication describes a transition from the previously used flat annotation scheme to a hierarchical structure for morphological tags (e.g., `V;PST;NOM(3,SG)`), the datasets currently distributed by the project remain annotated in the original flat format. Consequently, all data used in this thesis follows this flat annotation style.

UniMorph data served as the main resource throughout the experimental pipeline. As in the original SIGMORPHON 2023 Shared Task setup, it formed the basis for the reimplementation of the shared task baselines.

Furthermore, the dataset was used to fine-tune the ByT5 model for morphological inflection and, by inverting the source–target mapping, also for morphological analysis.

(a) UniMorph instance

Lemma	Tags	Inflected form
walk	V;PST;3;SG	walked

(b) Tag interpretation

Tag	Meaning
V	Verb
PST	Past tense
3	Third person
SG	Singular

Table 3.1: Example UniMorph instance in the tab-separated format used in this work (left), together with the interpretation of the corresponding morphological tags (right).

It additionally constituted the core training data for the bidirectional model, while certain experimental settings were further supplemented with data derived from Universal Dependencies.

3.1.2 Universal Dependencies

The UD resource provides consistent morphosyntactic annotation across languages, offering dependency treebanks with standardized part-of-speech tags, morphological features, and syntactic relations for sentences. At the time of writing, UD covers over 150 languages through more than 200 treebanks, spanning both high-resource and low-resource settings.

The data is distributed in the CoNLL-U format, where each sentence is represented as a sequence of token-level annotations. In addition to syntactic dependency information, each token is associated with a lemma and a set of morphological features encoded as attribute–value pairs (e.g., `Tense=Past|Number=Sing|Person=3`). Compared to UniMorph's tab-separated triple format, UD therefore provides richer contextual and sentence-level information alongside morphological annotations. An example taken from the Portuguese PetroGold treebank is shown in Figure 3.1.

```
# text = Estas podem representar falhas profundas.
# sent_id = 247-20140910-MONOGRAFIA_0-42
1 Estas este PRON _ Gender=Fem|Number=Plur|PronType=Dem 2 nsubj _ _
2 podem poder VERB _ Mood=Ind|Number=Plur|Person=3|Tense=Pres|VerbForm=Fin 0 root _ _
3 representar representar VERB _ VerbForm=Inf 2 xcomp _ _
4 falhas falha NOUN _ Gender=Fem|Number=Plur 3 obj _ _
5 profundas profundo ADJ _ Gender=Fem|Number=Plur 4 amod _ SpaceAfter=No
6 . . PUNCT _ _ 2 punct _ _
```

Figure 3.1: Example sentence in CoNLL-U format from the Portuguese UD PetroGold treebank ("These may represent deep faults.").

In this work, UD data serves as a source of contextual information for fine-tuning the bidirectional ByT5 model and for conducting experiments with large language models.

Since UD often provides multiple treebanks for a given language, it was necessary to select a single resource for each case. To guide this choice, we relied on the ranking provided by the UD project, which evaluates treebanks based on their size, annotation quality, and linguistic diversity. Whenever multiple treebanks were available for a language, the highest-ranked treebank according to this score was selected.

3.1.3 Data Preparation

Since the two data sources use different annotation schemes and file formats, several preprocessing and conversion steps were required to obtain a unified representation suitable for all experiments.

UniMorph. For UniMorph, the data was randomly sampled from the original datasets with a fixed seed, giving a preference to verbs, and then split into training, development, and test sets. This preference reflects that verbs typically exhibit the richest inflectional paradigms and constitute the primary focus of the morphological inflection task. These partitions consist of 10,000 training instances, 1,000 development instances, and 1,000 test instances, as in the Shared Task setup.

To prevent data leakage across splits, sampling and partitioning is performed at the lemma level, i.e., all inflected forms associated with a given lemma are always assigned to the same partition (Goldman et al., 2022). Verbal instances are identified using the UniMorph tag string, where entries whose feature bundle begins with `V` (including prefixed variants such as participles) are treated as verbs. If a split exceeds the target size, instances are subsampled within the split. Finally, all partitions are written in a unified tab-separated format (*lemma*, *tags*, *form*) to ensure consistency with the downstream baselines and model implementations.

While Table 3.3 reports the average number of inflected forms per lemma across the complete UniMorph datasets, this value varies substantially across parts of speech. In particular, Georgian exhibits a much richer verbal paradigm than suggested by the overall average. When considering verbs only, the average number of forms per lemma increases from approximately 24 to 178 forms. Because the data preparation procedure prioritizes verbal instances, this discrepancy has a noticeable impact on the effective data distribution and on the resulting train–development–test splits for Georgian.

To maintain comparable dataset sizes across languages, the preprocessing script allows specifying a maximum number of forms retained per lemma. For Georgian, a language-specific limit of 15 forms was applied in order to prevent individual paradigms from dominating the sampled data while preserving broad lemma coverage.

Universal Dependencies. For most UD treebanks, predefined training, development, and test splits are provided. However, for Amharic only a test set is available. In this case, this data was randomly partitioned into training, development, and test sets following an 8/1/1 proportion, as no official split specification is provided by UD and no common ratio can be derived from the analyzed languages.

To integrate UD data into the same downstream pipeline as UniMorph, all instances are converted into a unified tab-separated format that closely follows the UniMorph specification. Each instance consists of a lemma, a set of morphological tags, and the corresponding inflected form, separated by tab characters; for contextual settings, an additional fourth column containing the sentence context is included.

To incorporate UD data into this format, the morphological feature annotations provided in the CoNLL-U files were first mapped to their UniMorph equivalents. For this purpose, we used the official UniMorph conversion tool, which translates UD feature sets into the UniMorph schema using language-specific translators (McCarthy et al., 2018). Since Georgian and Amharic are not supported by the provided translators, the experiments requiring UD data for them were done with the original UD annotation style.

Figure 3.2 illustrates the output of the conversion tool when applied to the sentence shown in Figure 3.1. The UniMorph-compatible tags replace the original UD feature representations while preserving the original tokenization and syntactic structure.

```
# text = Estas podem representar falhas profundas.
# sent_id = 247-20140910-MONOGRAFIA_0-42
1  Estas  este  PRON  _  PRO;PL;FEM  2  nsubj  _  _
2  podem  poder  VERB  _  PL;PRS;V;IND;3  0  root  _  _
3  representar  representar  VERB  _  V;NFIN  2  xcomp  _  _
4  falhas  falha  NOUN  _  PL;FEM;N  3  obj  _  _
5  profundas  profundo  ADJ  _  PL;FEM;ADJ  4  amod  _  SpaceAfter=No
6  .  .  PUNCT  _  _  2  punct  _  _
```

Figure 3.2: Example output from official conversion tool.

Since the UniMorph data predominantly contains verbs, UD data was filtered accordingly to include only verbal tokens. Concretely, for each sentence, the first verb was selected and used to construct a single instance.

After conversion, the lemma, the inflected form, the mapped UniMorph tags, and the surrounding sentence context were extracted to construct the final tab-separated instances used in the downstream experiments. The resulting unified format is illustrated in Table 3.2.

All preprocessing, conversion, and extraction scripts are included in the accompanying code repository to ensure full reproducibility.

Lemma	Tags	Inflected form	Context
poder	V;IND;PRS;PL;3	podem	Estas podem representar falhas profundas.

Table 3.2: Example instance of the unified tab-separated format used in this work.

3.2 Languages

The following tables summarize the languages considered in the experiments. Table 3.3 presents general typological and linguistic information, together with the linguistic diversity score proposed by Joshi et al. (2020) as an external measure of language resource availability. In addition, the total number of UniMorph instances is reported to reflect the overall amount of available training data. The average number of inflected forms per lemma is included as a coarse estimate of morphological richness.

Table 3.4 provides statistics for the Universal Dependencies resources. Since UD treebanks are not available for Amuzgo and Lower Sorbian, these languages are excluded from the contextual and large language model experiments. For the remaining languages, the selected treebank and the sizes of the training, development, and test splits are listed; for Amharic, these correspond to the splits obtained by randomly partitioning the original UD test-only data into train/dev/test.

Language	ISO	Family	Type	Ling. score	UM size	Avg. forms
Amharic	amh	Semitic	Templatic	2	46224	18.78
Amuzgo	azg	Oto-Manguean	Polysynthetic	0	12204	36.76
Ancient Greek	grc	IE/Hellenic	Fusional	3	41593	18.10
Lower Sorbian	dsb	IE/Slavic	Fusional	1	20121	20.22
English	eng	IE/Germanic	Analytic	5	115523	1.63
Italian	ita	IE/Romance	Fusional	4	509574	50.91
Georgian	kat	Kartvelian	Agglutinative	3	74412	23.90
Portuguese	por	IE/Romance	Fusional	4	303996	75.96

Table 3.3: Typological overview and UniMorph statistics for the selected languages. Language identifiers follow ISO 639-3 and are used throughout the Results tables. For readability, Indo-European language families are prefixed with *IE* (Indo-European). *Ling. score* (0–5; higher means better represented in NLP resources and literature) provides a coarse external indicator of language resource availability. *UM size* denotes the number of UniMorph instances (lemma–tag bundle–inflected form triples) available for each language.

Language	Treebank	Train	Dev	Test
Amharic	ATT	811	102	99
Ancient Greek	PTNK	692	424	404
English	GUM	7692	1220	1171
Italian	ISDT	10728	475	404
Georgian	GNC	798	120	741
Portuguese	PetroGold	5871	568	894

Table 3.4: Universal Dependencies treebanks and dataset sizes for the languages with available UD resources.

4 Methodology

All experiments in this thesis are executed through a unified runner that separates implementation from configuration. Model- and language-specific settings are defined in YAML configuration files (including data paths, hyperparameters, and prompt templates), while a single entry point orchestrates training and inference across languages. To ensure consistent reporting across model families and tasks, each run writes a language-specific JSON file containing the evaluation metrics and relevant metadata, and additionally appends a summary row to a central CSV file aggregating all results. The full set of scripts, configurations, and evaluation utilities used to produce the reported results is made available in the accompanying repository to support complete reproducibility.

4.1 Morphological Inflection

This section describes how the different systems are trained and evaluated for morphological inflection. For each model, we summarize the concrete experimental procedure (data interface, training and inference setup, and implementation details relevant for reproducibility). Unless stated otherwise, all models operate on the unified UniMorph-based train–development–test splits described in Chapter 3.

4.1.1 Non-Neural Baseline

As a non-neural reference system, we use the official SIGMORPHON shared-task implementation (Goldman et al., 2023) described in Section 2.2.1. The baseline code is used as provided[1], with a minor command-line extension that allows restricting execution to a single language (via a `-lang` argument) for convenience and to simplify batched experiment runs.

4.1.2 Neural Baseline

As a neural reference system, we use the official SIGMORPHON 2023 neural transducer baseline (Goldman et al., 2023) described in Section 2.2.2. Training is performed by

[1] `https://github.com/sigmorphon/2023InflectionST/blob/main/part1/baselines/nonneural.py`

invoking the shared-task example script[2], which calls the baseline training code with a fixed configuration and takes the language code as an argument.

For each language, the model is trained on the corresponding training split, monitored on the development split, and evaluated on the test split. The shared-task training code produces a decoded prediction file in TSV format for the test set, which we use as model outputs when computing and reporting accuracy.

Table 4.1 summarizes the hyperparameters used in our experiments as specified by the shared-task script.

Hyperparameter	Value
Architecture	feature-invariant Transformer (`tagtransformer`)
Encoder / decoder layers	4 / 4
Embedding dimension	256
Hidden size (d_{FF} / internal)	1024
Attention heads	4
Dropout	0.3
Batch size	400
Learning rate	0.001
Scheduler	`warmupinvsqr` (warmup: 4000 steps)
Training budget	20000 update steps (`max_steps`)
Label smoothing	0.1
Adam β_2	0.98
Gradient clipping	max norm 1
Decoding	greedy, max decode length 32
Checkpoint selection	best development accuracy (`-bestacc`)

Table 4.1: Hyperparameters of the SIGMORPHON 2023 neural baseline as used in our experiments.

4.1.3 ByT5

For morphological inflection, we fine-tune a pretrained ByT5 model using the HuggingFace Transformers framework. Concretely, we load the `google/byt5-small` checkpoint via `AutoTokenizer` and `AutoModelForSeq2SeqLM` and train using PyTorch with the `Seq2SeqTrainer` API. Training is performed on GPU.

[2]https://github.com/omagolda/neural-transducer/tree/master/example/sigmorphon2023-shared-tasks

Each training instance is represented as a sequence-to-sequence example. The input string is constructed from the lemma and its UniMorph feature bundle using a fixed prompt template (Figure 4.1), and the target string is the corresponding inflected form. Inputs are processed with truncation to a maximum length of 128.

```
Generate the inflected form for:  {lemma} {features}
```

Figure 4.1: Prompt template used to construct the ByT5 input sequence for morphological inflection.

Table 4.2 summarizes the training and decoding hyperparameters used in our experiments.

Hyperparameter	Value
Pretrained checkpoint	`google/byt5-small`
Epochs	3
Learning rate	1×10^{-4}
Scheduler	linear (warmup: 100 steps)
Weight decay	0.01
Batch size (train / eval, per device)	8 / 8
Mixed precision	fp16 disabled
Seed	42
Evaluation / saving	every 50 / 100 steps (keep last 1 checkpoint)
Max input length (encoding)	128
Max target length (encoding)	128
Decoding	greedy
Max generation length (eval/test)	128
Prediction batch size	16

Table 4.2: ByT5 fine-tuning hyperparameters used for the morphological inflection experiments.

4.1.4 Evaluation

All models are evaluated on held-out test instances by comparing each generated output to the gold label provided in the corresponding dataset test split. For the morphological inflection task, the primary metric is exact-match accuracy on the inflected form, i.e., the fraction of test instances for which the predicted inflected form exactly matches the gold form. String comparisons are performed after applying Unicode normalization (NFC), lowercasing, and whitespace trimming.

In addition to accuracy, we report the mean Levenshtein distance between the predicted and gold inflected forms (computed on the same normalized strings) as a complementary measure that captures the magnitude of character-level deviations when the prediction is not exactly correct.

4.2 Morphological Analysis

This section describes the experimental setup for morphological analysis.

4.2.1 ByT5

For the analysis ByT5 model, we fine-tune ByT5 in the inverse direction on the UniMorph-based dataset. The only change relative to inflection is the input–output mapping and the corresponding prompt template: the model receives an inflected form as input and generates a single target string of the form "`lemma FEATURES`", where the lemma is followed by the semicolon-separated UniMorph tag bundle (separated by whitespace). We use the inverse prompt template shown in Figure 4.3.

Encoding and optimization hyperparameters are identical to the inflection ByT5 setup (Table 4.2).

4.2.2 Large Language Models

We evaluate large language models in an inference-only setting via the OpenRouter chat-completions API on the UD-derived unified dataset with context. Each instance is processed independently, i.e., every prediction is generated through a separate API request without retaining conversational history between examples. To reduce stochasticity and improve reproducibility, we fix the decoding configuration across all experiments by setting the temperature to 0.0 and by using a constant random seed.

We employ a one-shot prompting strategy. Each prompt consists of (i) a short instruction describing the task, (ii) a single labeled example drawn from the first data point of the development split, and (iii) the target instance for which a prediction is required. The same template and demonstration example are included in every request to ensure consistent behavior across instances. An example of the prompt template is shown in Figure 4.2.

```
Based on this example:
Input: {example_form}
Context: {example_context}
Prediction: {example_lemma}\t{example_tags}

Generate the lemma and morphological tags for the following inflected
verb and context. IMPORTANT: Provide ONLY the answer in the exact format
'lemma\ttags' with NO explanations, NO commentary, NO additional text:
{form}\t{context}

Answer (lemma and tags only, no explanation):
```

Figure 4.2: Prompt structure used for the one-shot LLM baseline. The target instance consists of an inflected form together with its sentence context.

4.2.3 Evaluation

For morphological analysis, the target output is a single string of the form "`lemma FEATURES`". The evaluation splits the prediction and the gold label at the first whitespace into a lemma and an MSD. The MSD component is interpreted as a case-insensitive set of semicolon-separated tags.

We report (i) lemma exact-match accuracy, and (ii) the mean Levenshtein distance between the predicted and gold lemmas (computed on normalized strings) to quantify character-level deviations. For MSD, we compute (iii) exact set-match accuracy (prediction set equals gold set) and (iv) micro-averaged precision, recall, and F_1 over tags, where true positives, false positives, and false negatives are accumulated across all test instances before computing the final scores.

For Georgian and Amharic in the UD-based context experiments, the available UD feature bundles cannot be converted to the UniMorph format due to missing language-specific translators. Since this yields feature inventories that are not directly comparable to the UniMorph-style MSD tags used for other languages, we do not report the MSD-based scores for these two languages, and instead report only the lemma metrics.

4.3 Bidirectional Inflection and Analysis

In addition to the task-specific ByT5 models, we fine-tune a single bidirectional model that jointly learns morphological inflection and morphological analysis within a unified sequence-to-sequence framework. Concretely, for each language we shuffle the Uni-

Morph training split with a fixed random seed and partition it into two equally sized subsets. One subset is converted into forward (inflection) examples (lemma + tags → inflected form) using the inflection prompt template (Figure 4.1), while the other subset is converted into inverse (analysis) examples (inflected form → lemma + tags) using a separate prompt template (Figure 4.3). The combined dataset is then used to fine-tune a single model.

The same procedure is applied to the development split to form a mixed-direction validation set. At test time, we evaluate the bidirectional model in the forward direction for morphological inflection and in the inverse direction for morphological analysis.

```
Generate the lemma and morphological tags for the following inflected
verb: {form}
```

Figure 4.3: Prompt template used to construct the ByT5 input sequence for the inverse mapping (morphological analysis) in the bidirectional setting.

Table 4.3 summarizes the hyperparameters used for bidirectional fine-tuning and decoding.

Hyperparameter	Value
Pretrained checkpoint	`google/byt5-small`
Training directions	**50% forward + 50% inverse (random split)**
Epochs	3
Learning rate	1×10^{-4}
Scheduler	linear (warmup: 100 steps)
Weight decay	0.01
Batch size (train / eval, per device)	8 / 8
Mixed precision	fp16 disabled
Seed	42
Evaluation / saving	every 50 / 100 steps (keep last 1 checkpoint)
Max input length (encoding)	128
Max target length (encoding)	128
Decoding	greedy
Max generation length (eval/test)	128
Prediction batch size	16

Table 4.3: ByT5 bidirectional fine-tuning hyperparameters used in our experiments.

4.4 Context-Aware Morphological Analysis

To incorporate sentence context for morphological analysis, we start from the language-specific bidirectional ByT5 model trained in Section 4.3 and further fine-tune it on the UD-derived unified dataset that provides sentence context (one model per language). Table 4.4 summarizes the key differences introduced by this additional fine-tuning stage relative to the standard bidirectional setup. All remaining optimization and decoding hyperparameters are taken unchanged from Table 4.3.

Hyperparameter	Value
Pretrained checkpoint	`results/byt5-bidir-{lang}/checkpoint-3750`
Inference direction	**inverse only**
Maximum context length	512

Table 4.4: Key differences between the context-aware model and the standard bidirectional ByT5 setup (Table 4.3).

At inference time, we decode only in the inverse (analysis) direction.

For each instance, we include the sentence context from the UD-derived dataset (the sentence in which the target inflected verb occurs) directly in the model input via the prompt. The resulting template is shown in Figure 4.4.

```
Given the context: '{context}'
Generate the lemma and morphological tags for the inflected verb:
{form}
```

Figure 4.4: Prompt template used for the context-aware ByT5 inverse model.

5 Results

5.1 Morphological Inflection

This section reports test-set performance on the morphological inflection task using the UniMorph-based datasets described in Chapter 3. Each column corresponds to one language, identified by its ISO code (see Table 3.3). Results are summarized in Table 5.1, which reports both accuracy and mean Levenshtein distance.

Model	amh	azg	dsb	eng	grc	ita	kat	por
Accuracy (%, ↑)								
Non-neural	39.0	6.1	**85.2**	95.8	29.0	76.2	84.2	96.6
Neural	**88.7**	20.1	79.1	96.6	19.4	78.3	**89.2**	**97.6**
ByT5	82.2	**26.5**	83.9	97.1	**41.6**	**95.8**	85.1	95.0
ByT5 bi	60.8	24.4	78.3	**97.6**	34.8	89.8	82.9	91.1
Mean Levenshtein distance (↓)								
Non-neural	1.010	5.660	**0.267**	0.086	1.852	0.761	0.430	0.054
Neural	**0.162**	3.703	0.379	0.069	2.212	0.266	**0.297**	**0.031**
ByT5	0.237	**3.413**	0.287	0.067	**1.713**	**0.091**	0.480	0.072
ByT5 bi	0.606	3.564	0.375	**0.050**	2.020	0.243	0.634	0.149

Table 5.1: Inflection results (test set): accuracy and mean Levenshtein distance.

Accuracy measures exact-match correctness of the predicted inflected form, while the mean Levenshtein distance captures the average character-level edit distance between predictions and gold forms (lower is better). Throughout this chapter, boldface indicates the best model for each language within each metric; in all languages in Table 5.1, the same model is best under both accuracy and mean Levenshtein distance.

Overall, the best-performing models are split between the neural baseline and the ByT5 variants, suggesting that both strong task-specific architectures and large pre-trained sequence-to-sequence models can be competitive depending on the language. Moreover, the bidirectional ByT5 variant is trained on a 50/50 mixture of forward and inverse UniMorph examples (Section 4.3), yet it remains competitive in the forward direction and even outperforms the neural baseline for several languages (e.g., azg, eng, grc, ita). A notable exception is Lower Sorbian (dsb), where the non-neural baseline achieves the best accuracy and the lowest mean Levenshtein distance, indicating that well-designed rule-based approaches can still match or outperform neural systems in some settings.

5.2 Morphological Analysis

This section reports test-set performance on morphological analysis. Each column corresponds to one language, identified by its ISO code (see Table 3.3). As in the evaluation pipeline, results are reported for (i) predicting the lemma and (ii) predicting the target MSD (interpreted as a set of tags). Since the inverse and bidirectional ByT5 models are trained and tested on UniMorph, while the context-based and LLM experiments use UD-derived data, results are reported separately for the two settings.

UniMorph-based results for the ByT5 inverse models are shown in Table 5.2 (lemma metrics) and Table 5.3 (MSD metrics).

Model	amh	azg	dsb	eng	grc	ita	kat	por
Lemma accuracy (%, ↑)								
ByT5 inv	**76.2**	34.4	61.3	**97.0**	59.7	**94.5**	75.6	**94.8**
ByT5 bi-inv	64.6	**37.1**	**64.3**	96.8	**64.0**	92.6	**76.2**	94.2
Lemma mean Levenshtein (↓)								
ByT5 inv	**0.287**	2.618	0.597	**0.044**	0.737	**0.099**	0.462	**0.076**
ByT5 bi-inv	0.408	**2.343**	**0.546**	0.048	**0.701**	0.141	**0.434**	0.091

Table 5.2: Morphological analysis (UniMorph, test set): lemma metrics.

For UniMorph-based analysis, the model that achieves the best lemma accuracy for a given language is also best under lemma mean Levenshtein distance (Table 5.2). Notably, the bidirectional inverse model (ByT5 bi-inv), which is trained on a 50/50 mix of forward and inverse examples (Section 4.3), achieves the best lemma results in exactly half of the languages (azg, dsb, grc, kat), indicating that splitting supervision across the two directions does not necessarily harm lemma prediction. For MSD prediction, ByT5 inv yields the strongest results in most languages; the winner is consistent between MSD accuracy and F1 in all but one case (kat), where the two metrics select different models (Table 5.3).

Model	amh	azg	dsb	eng	grc	ita	kat	por
MSD accuracy (%, ↑)								
ByT5 inv	**83.2**	**77.8**	**46.2**	81.3	**73.4**	**78.9**	72.7	**66.1**
ByT5 bi-inv	72.3	69.2	36.8	**81.7**	63.8	78.1	**73.0**	65.7
MSD F1 (%, ↑)								
ByT5 inv	**92.8**	**94.4**	**72.4**	96.4	**90.6**	**93.0**	**87.0**	**89.0**
ByT5 bi-inv	88.4	89.0	67.9	**96.6**	84.9	92.5	84.9	88.5

Table 5.3: Morphological analysis (UniMorph, test set): MSD metrics.

UD-based results for the context-based ByT5 model and the LLM baselines are shown in Table 5.4 (lemma metrics) and Table 5.5 (MSD metrics).

Model	amh	eng	grc	ita	kat	por
Lemma accuracy (%, ↑)						
ByT5 ctx	**89.9**	**96.9**	14.4	**95.5**	**6.1**	**74.5**
LLM Llama	66.7	66.4	**35.1**	78.2	4.9	71.3
LLM Qwen	47.5	49.1	32.7	42.8	4.0	51.6
Lemma mean Levenshtein (↓)						
ByT5 ctx	**0.131**	**0.072**	**4.030**	**0.149**	**3.188**	**0.596**
LLM Llama	0.798	0.855	4.057	2.002	4.750	0.751
LLM Qwen	4.323	5.465	6.077	8.332	17.819	8.453

Table 5.4: Morphological analysis (UD, test set): lemma metrics.

As discussed in Section 4.2.3, for the UD-based setting MSD metrics are only reported for languages whose UD-derived labels could be converted into the evaluation tag inventory; therefore, Amharic (`amh`) and Georgian (`kat`) are omitted from Table 5.5.

In the UD-based setting, the context-aware ByT5 model achieves the strongest performance for most languages and metrics (Tables 5.4 and 5.5), substantially outperforming the prompted LLM baselines. Ancient Greek is a notable exception: for `grc`, the LLM baselines outperform the context-aware model on lemma accuracy and, even more clearly, on MSD accuracy and F1. At the same time, the lemma mean Levenshtein distances for `grc` remain high across all models, suggesting that the UD-derived Greek setting is particularly challenging in this configuration.

Two dataset-specific effects are worth noting when interpreting language-level outliers. First, in the prepared UniMorph subset used in this thesis, the Ancient Greek

Model	eng	grc	ita	por
MSD accuracy (%, ↑)				
ByT5 ctx	**95.7**	13.6	**96.8**	**71.8**
LLM Llama	10.5	36.6	3.5	1.9
LLM Qwen	6.7	**47.8**	21.8	2.8
MSD F1 (%, ↑)				
ByT5 ctx	**98.3**	30.6	**98.4**	**83.8**
LLM Llama	37.5	81.7	17.7	45.8
LLM Qwen	40.7	**81.8**	24.7	41.5

Table 5.5: Morphological analysis (UD, test set): MSD metrics.

(`grc`) splits contain nominal paradigms only (nouns and adjectives; no verb instances), whereas the UD-derived analysis experiments are verb-focused; consequently, Greek results are not directly comparable across the two data sources. This mismatch is a plausible contributing factor to why `grc` behaves differently in the UD-based tables, including the observation that the LLM baselines outperform the context-aware ByT5 model for Greek, despite the strong UniMorph-based results for ByT5 on `grc`. Second, Georgian exhibits a systematic mismatch in lemma conventions between UniMorph and UD: in the UniMorph Georgian data, verbs are grouped under a UniMorph-specific citation form (see also (Guriel et al., 2022)), while UD lemmas for Georgian are typically based on the *masdar* (verbal noun) form.[1] This difference can substantially affect lemma-level scores and makes cross-source comparisons for Georgian particularly sensitive to annotation conventions.

[1]UD lemmatization convention is described in the UD_Georgian-GNC README: `https://github.com/UniversalDependencies/UD_Georgian-GNC/blob/master/README.md`.

6 Conclusion

This thesis investigated how different modeling approaches perform on two fundamental tasks in morphological processing: (i) morphological inflection (lemma + feature bundle $\rightarrow$ inflected form) and (ii) morphological analysis (inflected form $\rightarrow$ lemma + MSD). Experiments were conducted across a set of typologically diverse languages using both UniMorph and UD-derived data, with the overarching goal of obtaining a clear, reproducible comparison under a unified preprocessing and evaluation pipeline.

Summary of approaches. For inflection, we evaluated a non-neural shared-task baseline, a neural transducer baseline, and fine-tuned ByT5 models, including a bidirectional variant trained on a mixture of forward and inverse examples. For analysis, we evaluated ByT5 models trained on UniMorph in the inverse direction and compared them to UD-based context experiments, including a context-aware ByT5 model and prompting-based LLM baselines.

Key findings. Across languages, fine-tuned sequence-to-sequence models (ByT5) provide a strong and generally robust approach for morphological generation and inversion, especially when trained directly on the target task and annotation scheme. At the same time, performance varies substantially by language, reflecting differences in paradigm size, orthographic complexity, and effective training-set coverage.

For morphological analysis, the results support a clear separation between (i) UniMorph-trained inverse models and (ii) UD-derived context and LLM experiments. UniMorph-trained models benefit from a consistent tag inventory and a direct supervision signal for the UniMorph-style MSD representation. In contrast, UD-based experiments introduce an additional conversion layer from UD feature bundles to UniMorph-style tags. When conversion is incomplete or not available, comparability breaks down for MSD metrics, motivating the reporting choices described in Section 4.2.3.

Finally, several language-level outliers illustrate that "better" or "worse" numbers can arise from dataset and annotation differences rather than from modeling alone. In particular, cross-resource comparisons are sensitive to part-of-speech coverage and to lemma conventions (e.g., verbal-noun versus finite-form lemmas), which can strongly impact lemma-level scores even when the underlying model behavior is stable.

Limitations. This work is limited by (i) the fact that, for the specific set of languages considered in this thesis, UniMorph and UD resources were not always available in a directly comparable form, (ii) the reliance on a fixed set of hyperparameters for fine-tuning, and (iii) the need to align resources with different annotation schemes. The LLM evaluation is additionally constrained by prompt design choices and by the difficulty of eliciting structured outputs reliably without task-specific fine-tuning.

Future work. Several extensions would strengthen the conclusions and broaden applicability. First, future work could conduct broader hyperparameter sweeps for ByT5 fine-tuning (learning rate schedules, sequence lengths, regularization, and decoding), and evaluate larger pretrained checkpoints to test scaling effects. Second, it would be valuable to expand to additional languages with both UniMorph and UD resources, prioritizing settings where UD→UniMorph translation is available or can be developed.

Finally, improved cross-resource evaluation could reduce confounds: rather than forcing all UD experiments into a UniMorph-style MSD space, one could report results in both tag inventories (or develop more faithful translators) to separate modeling limitations from conversion artifacts.

In summary, the results in this thesis indicate that modern sequence-to-sequence models are effective tools for morphological inflection and analysis across diverse languages, but that careful attention to dataset construction and annotation conventions is essential for drawing reliable cross-language and cross-resource conclusions.

Abbreviations

UD Universal Dependencies

MSD morphosyntactic description

NLP natural language processing

LLM large language model

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